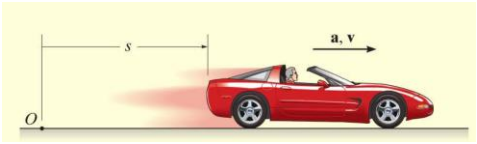
TUTORIAL 1: KINEMATICS OF A PARTICLE

ME 1060: DYNAMICS

Question 1

A car is moving in a straight line such that for a short time its velocity is defined by $v = (3t^2 + 2t)\ ft/s$ where t is in seconds. Determine its position and acceleration when $t = 3s$. When $t = 0$, $s = 0$.



Solution

Position

Since $v = f(t)$, the car's position can be determined from $v = \frac{ds}{dt}$.

Note that $s = 0$ when $t = 0$

We have $v = \frac{ds}{dt} = (3t^2 + 2t)$

$$\int_0^s ds = \int_0^t (3t^2 + 2t)\, dt$$

$$s|_0^s = t^3 + t^2|_0^t$$

$$s = t^3 + t^2$$

When $t = 3s$,

$$s = 3^3 + 3^2 = 36\ ft$$

Acceleration

$$a = \frac{dv}{dt} = \frac{d(3t^2 + 2t)}{dt}$$

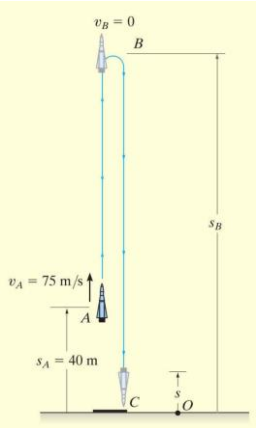
$$a = 6t + 2$$

When $t = 3s$,

$$a = 6(3) + 2 = 20\ ft/s^2$$

Question 2

During a test, a rocket travels upward at $75\ m/s$, and when it is $40\ m$ from the ground its engine fails. Determine the maximum height S_B reached by the rocket and its speed just before it hits the ground. While in motion the rocket is subjected to a constant downward acceleration of $9.81\ m/s^2$ due to gravity. Neglect the effect of air resistance.



Solution

Constant Acceleration $a = a_c$

When the acceleration is a constant where $a = a_c$, each of the three kinematic equations $a_c = dv/dt$, $v = ds/dt$, and $a_c\, ds = v\, dv$ can be integrated to obtain formulas that relate a_c, v, s, and t.

Velocity as a function of time:

Integrate $a_c = dv/dt$, assuming that initially $v = v_0$ when $t = 0$.

$$\int_{v_0}^{v} dv = \int_0^t a_c dt$$

$$v = v_0 + a_c t$$

Position as a function of time

Integrate $v = ds/dt = v_0 + a_c t$, assuming that initially $s = s_0$ when $t = 0$.

$$\int_{s_0}^{s} ds = \int_0^t (v_0 + a_c t) dt$$

$$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$$

Velocity as a function of position

integrate $v\, dv = a_c\, ds$, assuming that initially $v = v_0$ at $s = s_0$.

$$\int_{v_0}^{v} v dv = \int_{s_0}^{s} a_c ds$$

$$v^2 = v_0^2 + 2a_c(s - s_0)$$

Maximum Height

Since the rocket is traveling upward, $v_A = +75m/s$ when $t = 0$. At the maximum height $S = S_B$ the velocity $V_B = 0$. For the entire motion, the acceleration is $a_c = -9.81\ m/s^2$ (negative since it acts in the opposite sense to positive velocity or positive displacement). Since a_c is constant the rocket's position may be related to its velocity at the two points A and B on the path by using

$$v_B^2 = v_A^2 + 2a_c(s_B - s_A)$$

$$0 = 75^2 + 2(-9.81)(s_B - 40)$$

$$S_B = 327m$$

Velocity

To obtain the velocity of the rocket just before it hits the ground, we can apply

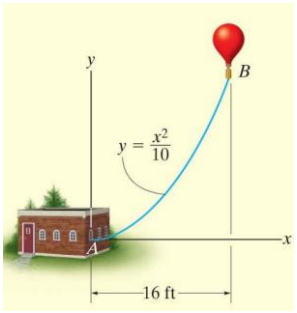
$$v_C^2 = v_B^2 + 2a_c(s_C - s_B)$$

$$v_c^2 = 0 + 2(-9.81)(0 - 327)$$

$$v_c = -80.1m/s = 80.1m/s \downarrow$$

Question 3

At any instant the horizontal position of the weather balloon in the figure is defined by $x = (8t)\ ft$, where t is in seconds. If the equation of the path is $y = x^2/10$, determine the magnitude and direction of the velocity and the acceleration when $t = 2\ s$.



Solution

Velocity

The velocity component in the x direction is

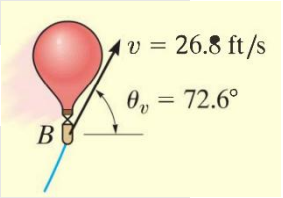
$$v_x = \dot{x} = \frac{d(8t)}{dt} = 8\ ft/s$$

To find the relationship between the velocity components we will use the chain rule of calculus.

$$v_y = \dot{y} = \frac{d(x^2/10)}{dt} = \frac{2x\dot{x}}{10} = 2(16)(8)/10 = 25.6\ ft/s \uparrow$$

When $t = 2\ s$, the magnitude of the velocity is therefore

$$v = \sqrt{(8\ fts^{-1})^2 + (25\ fts^{-1})^2} = 26.8\ fts^{-1}$$



The direction is tangent to the path where,

$$\theta_v = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\frac{25.6}{8} = 72.6^o$$

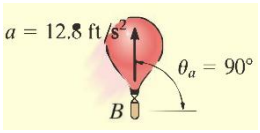
Acceleration

The relationship between the acceleration components is determined using the chain rule.

$$a_x = \dot{v}_x = \frac{d(8)}{dt} = 0$$

$$a_y = \dot{v}_y = \frac{d(2x\dot{x}/10)}{dt} = 2(\dot{x})\dot{x}/10 + 2x\ddot{x}/10$$

$$2(8)^2/10 + 2(16)(0)/10 = 12.8\ fts^{-2}$$



Thus,

$$a = \sqrt{(0)^2 + (12.8)^2} = 12.8\ fts^{-2}$$

The direction of a is

$$\theta_a = \tan^{-1}\frac{12.8}{0} = 90^o$$

Question 4

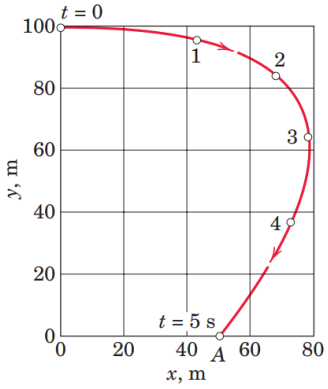
The curvilinear motion of a particle is defined by $v_x = 50 - 16t$ and $y = 100 - 4t^2$, where v_x is in meters per second, y is in meters, and t is in seconds. It is also known that $x = 0$ when $t = 0$. Plot the path of the particle and determine its velocity and acceleration when the position $y = 0$ is reached.

Solution

The x-coordinate is obtained by integrating the expression for v_x, and the x-component of the acceleration is obtained by differentiating v_x. Thus,

$$\left[\int dx = \int v_x dt\right] = \int_0^x dx = \int_0^t (50 - 16t)\, dt = 50t - 8t^2\ m$$

$$[a_x = \dot{v}_x] \qquad a_x = \frac{d(50 - 16t)}{dt} = -16 m/s^2$$

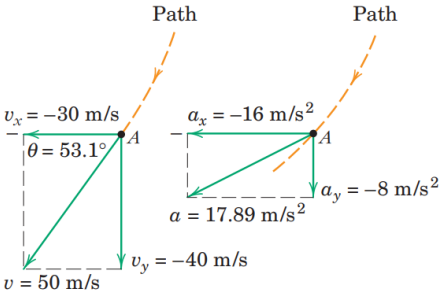


The y-components of velocity and acceleration are

$$[v_y = \dot{y}] \qquad v_y = \frac{d}{dt}(100 - 4t^2) = -8t\ \ m/s$$

$$[a_y = \dot{v}_y] \qquad a_y = \frac{d}{dt}(-8t) = -8\ m/s^2$$

We now calculate corresponding values of x and y for various values of t and plot x against y to obtain the path as shown.



When $y = 0$, $0 = 100 - 4t^2$. Therefore $t = 5s$. For this value of the time, we have,

$$v_x = 50 - 16(5) = -30\ m/s$$

$$v_y = -8(5) = -40\ m/s$$

$$v = \sqrt{(-30)^2 + (-40)^2} = 50\ \ m/s$$

$$a = \sqrt{(-16)^2 + (-8)^2} = 17.89\ \ m/s^2$$

The velocity and acceleration components and their resultants are shown on the separate diagrams for point A, where $y = 0$. Thus, for this condition we may write

$$\boldsymbol{v} = -30\boldsymbol{i} - 40\boldsymbol{j}\ \ m/s$$

$$\boldsymbol{a} = -16\boldsymbol{i} - 8\boldsymbol{j}\ m/s^2$$